

Lab Assignment 5

MTH 308: Numerical Analysis and Scientific Computing-I

Department of Mathematics and Statistics, IIT Kanpur

2026 Even Semester: February 20, 2026

This lab requires you to implement the Jacobi, Gauss-Seidel, and Successive Over-Relaxation (SOR) methods to approximate the solutions of linear systems. Please write a single script file named `Lab5LastnameRollNumber.m` and publish it as a PDF. Then submit the script `Lab5LastnameRollNumber.m` along with the corresponding published file `Lab5LastnameRollNumber.pdf` (for example, my submitted files would be `Lab5Biswas260102.m` and `Lab5Biswas260102.pdf`) to HELLO IITK.

Each solution should:

- Be contained in a separate cell, including the problem number and a short problem description.
- Be adequately commented to explain your logic.
- Run independently of other cells.

Write MATLAB functions to implement the Jacobi method, the Gauss-Seidel method, and the SOR method. These functions should be included at the end of the script as local functions. Each function should accept the inputs as the matrix A and right-hand side vector b , the maximum number of iterations, the stopping criterion tolerance, and the initial approximation vector $x^{(0)}$. In addition, the SOR function should accept one extra parameter, the relaxation parameter ω . Each function should return the number of iterations required, and the approximate solution vector. Output the number of iterations required and the approximate solution for each problem. Display at least k digits when the solution is accurate to within 10^{-k} .

Use the stopping criterion in terms of relative residual error:

$$\frac{\|b - Ax\|_2}{\|b\|_2} < \text{tolerance}.$$

The infinity norm $\|x\|_\infty$ can be computed in MATLAB using `norm(x, inf)`.

1. Consider the linear system:

$$\begin{array}{rrrrr} 10x_1 - & x_2 + & 2x_3 & & = 6 \\ -x_1 + 11x_2 - & x_3 + 3x_4 & & = 25 \\ 2x_1 - & x_2 + 10x_3 - & x_4 & = -11 \\ & 3x_2 - & x_3 + 8x_4 & = 15 \end{array}$$

- Use $x^{(0)} = \mathbf{0}$ and a tolerance of 10^{-4} to solve the system using the Jacobi method and the Gauss-Seidel method. Compare the number of iterations required for convergence by each method.
- Compute the spectral radii of the corresponding iteration matrices $\rho(M_J)$ and $\rho(M_{GS})$.

2. Consider the linear system:

$$\begin{aligned}
 4x_1 - x_2 &= 0 \\
 -x_1 + 4x_2 - x_3 &= 5 \\
 -x_2 + 4x_3 &= 0 \\
 &+ 4x_4 - x_5 = 6 \\
 -x_4 + 4x_5 - x_6 &= -2 \\
 -x_5 + 4x_6 &= 6
 \end{aligned}$$

Use $x^{(0)} = \mathbf{0}$ and a tolerance of 10^{-8} to solve the system using the Jacobi method, the Gauss-Seidel method, and the SOR method with four values of the relaxation parameter $\omega = 1.01$, $\omega = 1.05$, $\omega = 1.10$, $\omega = 1.15$, and $\omega = \frac{2}{1+\sqrt{1-\rho(M_J)^2}}$. Compare the number of iterations required for convergence by each method and conclude your observations.

Remark 1. Note that the coefficient matrix is symmetric positive definite with a block tridiagonal structure. For matrices of this type, the optimal SOR parameter is given by $\omega_{\text{opt}} = \frac{2}{1+\sqrt{1-\rho(M_J)^2}}$.